TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders

Standard → slope-intercept: subtract the x-term, then divide every term by the number with y.

Point-slope → slope-intercept: distribute, then add or subtract to get y alone.

Intercepts: x-intercept: put 0 in for y. y-intercept: put 0 in for x.

Solve a formula for a letter: undo everything around that letter, doing the same to both sides.

1. The formula for the area of a rectangle is $A = \ell w$. Solve the equation for w.

$w = A/\ell$

Divide both sides by ℓ

2. The formula for the volume of a cylinder is $V = \pi r^2 h$. Solve the equation for h.

h = V/(πr²)

Divide both sides by πr^2

3. Which equation is $d = rt$ solved for r?

A. $r = dt$

B. ***r = d/t***

C. $r = t/d$

D. $r = d - t$

Divide both sides by t

4. The lateral surface area of a can is $S = 2\pi rh$. $S = 37.68$ square inches and $r = 1.5$ inches. Use 3.14 for π. What is h?

h = 4 in.

$h = S/(2\pi r) = 37.68 \div 9.42 = 4$